

class13

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Section 1. Differential Expression Analysis

```
library(DESeq2)
```

Loading required package: S4Vectors

Warning: package 'S4Vectors' was built under R version 4.5.3

Loading required package: stats4

Loading required package: BiocGenerics

Loading required package: generics

Attaching package: 'generics'

The following objects are masked from 'package:base':

```
as.difftime, as.factor, as.ordered, intersect, is.element, setdiff,  
setequal, union
```

Attaching package: 'BiocGenerics'

The following objects are masked from 'package:stats':

```
IQR, mad, sd, var, xtabs
```

The following objects are masked from 'package:base':

```
anyDuplicated, aperm, append, as.data.frame, basename, cbind,  
colnames, dirname, do.call, duplicated, eval, evalq, Filter, Find,  
get, grep, grepl, is.unsorted, lapply, Map, mapply, match, mget,  
order, paste, pmax, pmax.int, pmin, pmin.int, Position, rank,  
rbind, Reduce, rownames, sapply, saveRDS, table, tapply, unique,  
unsplit, which.max, which.min
```

Attaching package: 'S4Vectors'

The following object is masked from 'package:utils':

```
findMatches
```

The following objects are masked from 'package:base':

```
expand.grid, I, unname
```

Loading required package: IRanges

Attaching package: 'IRanges'

The following object is masked from 'package:grDevices':

```
windows
```

Loading required package: GenomicRanges

Loading required package: Seqinfo

Loading required package: SummarizedExperiment

Loading required package: MatrixGenerics

Loading required package: matrixStats

Attaching package: 'MatrixGenerics'

The following objects are masked from 'package:matrixStats':

colAlls, colAnyNAs, colAnys, colAvgPerRowSet, colCollapse,
colCounts, colCummaxs, colCummins, colCumprods, colCumsums,
colDiffs, colIQRDiffs, colIQRs, colLogSumExps, colMadDiffs,
colMads, colMaxs, colMeans2, colMedians, colMins, colOrderStats,
colProds, colQuantiles, colRanges, colRanks, colSdDiffs, colSds,
colSums2, colTabulates, colVarDiffs, colVars, colWeightedMads,
colWeightedMeans, colWeightedMedians, colWeightedSds,
colWeightedVars, rowAlls, rowAnyNAs, rowAnys, rowAvgPerColSet,
rowCollapse, rowCounts, rowCummaxs, rowCummins, rowCumprods,
rowCumsums, rowDiffs, rowIQRDiffs, rowIQRs, rowLogSumExps,
rowMadDiffs, rowMads, rowMaxs, rowMeans2, rowMedians, rowMins,
rowOrderStats, rowProds, rowQuantiles, rowRanges, rowRanks,
rowSdDiffs, rowSds, rowSums2, rowTabulates, rowVarDiffs, rowVars,
rowWeightedMads, rowWeightedMeans, rowWeightedMedians,
rowWeightedSds, rowWeightedVars

Loading required package: Biobase

Welcome to Bioconductor

Vignettes contain introductory material; view with
'browseVignettes()'. To cite Bioconductor, see
'citation("Biobase")', and for packages 'citation("pkgname")'.

Attaching package: 'Biobase'

The following object is masked from 'package:MatrixGenerics':

rowMedians

The following objects are masked from 'package:matrixStats':

anyMissing, rowMedians

```
metaFile <- "GSE37704_metadata.csv"
countFile <- "GSE37704_featurecounts.csv"

# Import metadata and take a peek
colData = read.csv(metaFile, row.names=1)
head(colData)
```

```
      condition
SRR493366 control_sirna
SRR493367 control_sirna
SRR493368 control_sirna
SRR493369      hoxa1_kd
SRR493370      hoxa1_kd
SRR493371      hoxa1_kd
```

```
# Import countdata
countData = read.csv(countFile, row.names=1)
head(countData)
```

```
      length SRR493366 SRR493367 SRR493368 SRR493369 SRR493370
ENSG00000186092    918         0         0         0         0         0
ENSG00000279928    718         0         0         0         0         0
ENSG00000279457   1982        23        28        29        29        28
ENSG00000278566    939         0         0         0         0         0
ENSG00000273547    939         0         0         0         0         0
ENSG00000187634   3214       124       123       205       207       212
      SRR493371
ENSG00000186092         0
ENSG00000279928         0
ENSG00000279457        46
ENSG00000278566         0
ENSG00000273547         0
ENSG00000187634       258
```

Q. Complete the code below to remove the troublesome first column from `countData`

```
countData <- as.matrix(countData[, -1])
head(countData)
```

```
SRR493366 SRR493367 SRR493368 SRR493369 SRR493370 SRR493371
```

ENSG00000186092	0	0	0	0	0	0
ENSG00000279928	0	0	0	0	0	0
ENSG00000279457	23	28	29	29	28	46
ENSG00000278566	0	0	0	0	0	0
ENSG00000273547	0	0	0	0	0	0
ENSG00000187634	124	123	205	207	212	258

`countData[, -1]` means “all rows, all columns except the first.” The length column is gene length metadata, not expression counts.

Q. Complete the code below to filter `countData` to exclude genes (i.e. rows) where we have 0 read count across all samples (i.e. columns).

Tip: What will `rowSums()` of `countData` return and how could you use it in this context?

```
countData <- countData[rowSums(countData) > 1, ]
head(countData)
```

	SRR493366	SRR493367	SRR493368	SRR493369	SRR493370	SRR493371
ENSG00000279457	23	28	29	29	28	46
ENSG00000187634	124	123	205	207	212	258
ENSG00000188976	1637	1831	2383	1226	1326	1504
ENSG00000187961	120	153	180	236	255	357
ENSG00000187583	24	48	65	44	48	64
ENSG00000187642	4	9	16	14	16	16

`rowSums(countData) > 0` creates a logical vector that is TRUE for genes with at least one read across all samples

Running DESeq2

```
dds <- DESeqDataSetFromMatrix(countData = countData,
                              colData = colData,
                              design = ~condition)
```

Warning in `DESeqDataSet(se, design = design, ignoreRank)`: some variables in design formula are characters, converting to factors

```
dds <- DESeq(dds)
```

estimating size factors

estimating dispersions

gene-wise dispersion estimates

mean-dispersion relationship

final dispersion estimates

fitting model and testing

```
dds
```

```
class: DESeqDataSet
dim: 15280 6
metadata(1): version
assays(4): counts mu H cooks
rownames(15280): ENSG00000279457 ENSG00000187634 ... ENSG00000276345
               ENSG00000271254
rowData names(22): baseMean baseVar ... deviance maxCooks
colnames(6): SRR493366 SRR493367 ... SRR493370 SRR493371
colData names(2): condition sizeFactor
```

Q. Call the `summary()` function on your results to get a sense of how many genes are up or down-regulated at the default 0.1 p-value cutoff.

```
res <- results(dds)
summary(res)
```

```
out of 15280 with nonzero total read count
adjusted p-value < 0.1
LFC > 0 (up)           : 4351, 28%
LFC < 0 (down)         : 4399, 29%
outliers [1]           : 0, 0%
low counts [2]         : 590, 3.9%
(mean count < 1)
[1] see 'cooksCutoff' argument of ?results
[2] see 'independentFiltering' argument of ?results
```

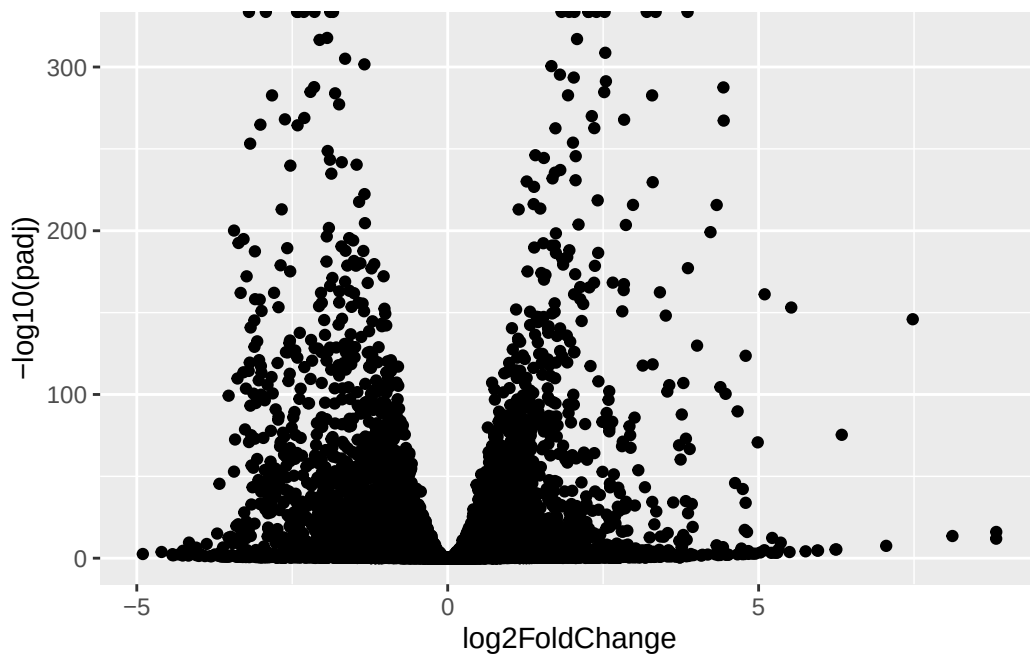
Volcano Plot

```
library(ggplot2)
```

Warning: package 'ggplot2' was built under R version 4.5.3

```
ggplot(as.data.frame(res)) +  
  aes(log2FoldChange, -log10(padj)) +  
  geom_point()
```

Warning: Removed 590 rows containing missing values or values outside the scale range (`geom\_point()`).



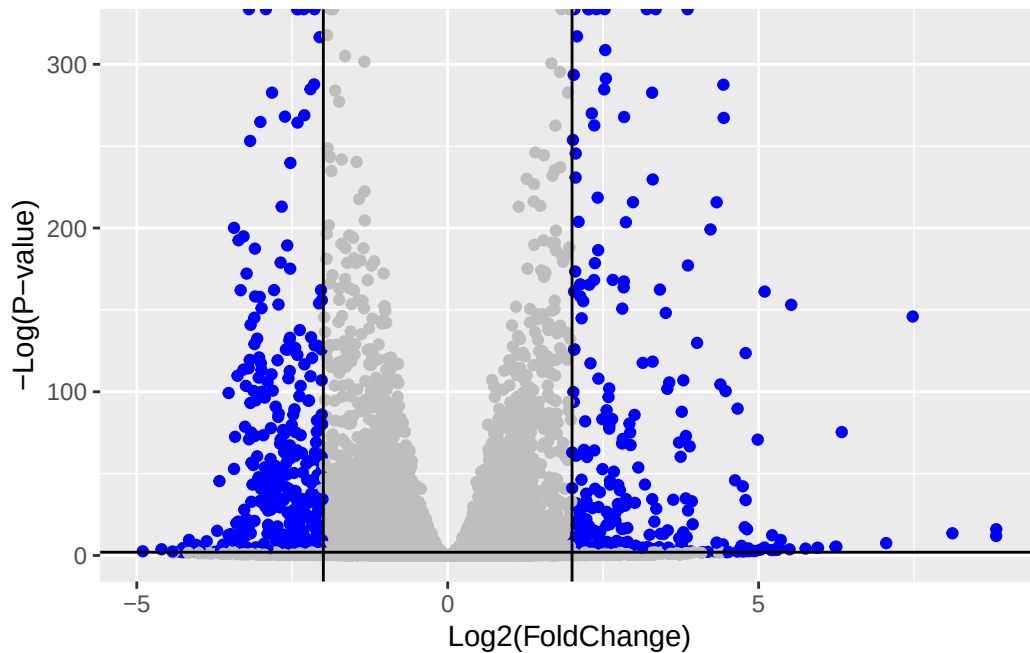
Q. Improve this plot by completing the below code, which adds color, axis labels and cutoff lines:

```
# Make a color vector for all genes  
mycols <- rep("gray", nrow(res))  
  
# Color blue the genes with absolute fold change above 2  
mycols[abs(res$log2FoldChange) > 2] <- "blue"  
  
# Color back to gray those with adjusted p-value more than 0.01
```

```
mycols[res$padj > 0.01] <- "gray"

ggplot(as.data.frame(res)) +
  aes(log2FoldChange, -log10(padj)) +
  geom_point(col = mycols) +
  xlab("Log2(FoldChange)") +
  ylab("-Log(P-value)") +
  geom_vline(xintercept = c(-2, 2)) +
  geom_hline(yintercept = -log10(0.01))
```

Warning: Removed 590 rows containing missing values or values outside the scale range (`geom\_point()`).



Adding gene annotation

Q. Use the `mapIDs()` function multiple times to add SYMBOL, ENTREZID and GENENAME annotation to our results by completing the code below.

```
library("AnnotationDbi")
library("org.Hs.eg.db")
```



```
res$symbol <- mapIds(org.Hs.eg.db,
                     keys = row.names(res),
                     keytype = "ENSEMBL",
                     column = "SYMBOL",
                     multiVals = "first")
```

'select()' returned 1:many mapping between keys and columns

```
res$entrez <- mapIds(org.Hs.eg.db,
                     keys = row.names(res),
                     keytype = "ENSEMBL",
                     column = "ENTREZID",
                     multiVals = "first")
```

'select()' returned 1:many mapping between keys and columns

```
res$name <- mapIds(org.Hs.eg.db,
                   keys = row.names(res),
                   keytype = "ENSEMBL",
                   column = "GENENAME",
                   multiVals = "first")
```

'select()' returned 1:many mapping between keys and columns

```
head(res, 10)
```

log2 fold change (MLE): condition hoxa1 kd vs control sirna

Wald test p-value: condition hoxa1 kd vs control sirna

DataFrame with 10 rows and 9 columns

	baseMean	log2FoldChange	lfcSE	stat	pvalue
	<numeric>	<numeric>	<numeric>	<numeric>	<numeric>
ENSG00000279457	29.9136	0.1792571	0.3248216	0.551863	5.81042e-01
ENSG00000187634	183.2296	0.4264571	0.1402658	3.040350	2.36304e-03
ENSG00000188976	1651.1881	-0.6927205	0.0548465	-12.630158	1.43990e-36
ENSG00000187961	209.6379	0.7297556	0.1318599	5.534326	3.12428e-08
ENSG00000187583	47.2551	0.0405765	0.2718928	0.149237	8.81366e-01
ENSG00000187642	11.9798	0.5428105	0.5215598	1.040744	2.97994e-01
ENSG00000188290	108.9221	2.0570638	0.1969053	10.446970	1.51282e-25

ENSG00000187608	350.7169	0.2573837	0.1027266	2.505522	1.22271e-02
ENSG00000188157	9128.4394	0.3899088	0.0467163	8.346304	7.04321e-17
ENSG00000131591	156.4791	0.1965923	0.1456109	1.350121	1.76977e-01
	padj	symbol	entrez		name
	<numeric>	<character>	<character>		<character>
ENSG00000279457	6.85033e-01	NA	NA		NA
ENSG00000187634	5.14039e-03	SAMD11	148398	sterile alpha motif	..
ENSG00000188976	1.75974e-35	NOC2L	26155	NOC2 like nucleolar	..
ENSG00000187961	1.13044e-07	KLHL17	339451	kelch like family me..	
ENSG00000187583	9.19159e-01	PLEKHN1	84069	pleckstrin homology	..
ENSG00000187642	4.02066e-01	PERM1	84808	PPARGC1 and ESRR ind..	
ENSG00000188290	1.30113e-24	HES4	57801	hes family bHLH tran..	
ENSG00000187608	2.36679e-02	ISG15	9636	ISG15 ubiquitin like..	
ENSG00000188157	4.20589e-16	AGRN	375790		agrin
ENSG00000131591	2.60893e-01	C1orf159	54991	chromosome 1 open re..	

Q. Finally for this section let's reorder these results by adjusted p-value and save them to a CSV file in your current project directory.

```
res <- res[order(res$pvalue), ]
write.csv(as.data.frame(res), file = "deseq_results.csv")
```

Section 2. Pathway Analysis

KEGG pathways

```
# One-time install (run in console, not in .qmd):
# BiocManager::install(c("pathview", "gage", "gageData"))

library(pathview)
```

```
#####
Pathview is an open source software package distributed under GNU General
Public License version 3 (GPLv3). Details of GPLv3 is available at
http://www.gnu.org/licenses/gpl-3.0.html. Particullary, users are required to
formally cite the original Pathview paper (not just mention it) in publications
or products. For details, do citation("pathview") within R.
```

The pathview downloads and uses KEGG data. Non-academic uses may require a KEGG license agreement (details at <http://www.kegg.jp/kegg/legal.html>).

```
#####
```

```
library(gage)
```

```
library(gageData)

data(kegg.sets.hs)
data(sigmet.idx.hs)

# Focus on signaling and metabolic pathways only
kegg.sets.hs <- kegg.sets.hs[sigmet.idx.hs]

# Examine the first 3 pathways
head(kegg.sets.hs, 3)
```

```
$`hsa00232 Caffeine metabolism`
```

```
[1] "10" "1544" "1548" "1549" "1553" "7498" "9"
```

```
$`hsa00983 Drug metabolism - other enzymes`
```

```
[1] "10" "1066" "10720" "10941" "151531" "1548" "1549" "1551"
[9] "1553" "1576" "1577" "1806" "1807" "1890" "221223" "2990"
[17] "3251" "3614" "3615" "3704" "51733" "54490" "54575" "54576"
[25] "54577" "54578" "54579" "54600" "54657" "54658" "54659" "54963"
[33] "574537" "64816" "7083" "7084" "7172" "7363" "7364" "7365"
[41] "7366" "7367" "7371" "7372" "7378" "7498" "79799" "83549"
[49] "8824" "8833" "9" "978"
```

```
$`hsa00230 Purine metabolism`
```

```
[1] "100" "10201" "10606" "10621" "10622" "10623" "107" "10714"
[9] "108" "10846" "109" "111" "11128" "11164" "112" "113"
[17] "114" "115" "122481" "122622" "124583" "132" "158" "159"
[25] "1633" "171568" "1716" "196883" "203" "204" "205" "221823"
[33] "2272" "22978" "23649" "246721" "25885" "2618" "26289" "270"
[41] "271" "27115" "272" "2766" "2977" "2982" "2983" "2984"
[49] "2986" "2987" "29922" "3000" "30833" "30834" "318" "3251"
[57] "353" "3614" "3615" "3704" "377841" "471" "4830" "4831"
[65] "4832" "4833" "4860" "4881" "4882" "4907" "50484" "50940"
[73] "51082" "51251" "51292" "5136" "5137" "5138" "5139" "5140"
[81] "5141" "5142" "5143" "5144" "5145" "5146" "5147" "5148"
[89] "5149" "5150" "5151" "5152" "5153" "5158" "5167" "5169"
[97] "51728" "5198" "5236" "5313" "5315" "53343" "54107" "5422"
```

```
[105] "5424" "5425" "5426" "5427" "5430" "5431" "5432" "5433"
[113] "5434" "5435" "5436" "5437" "5438" "5439" "5440" "5441"
[121] "5471" "548644" "55276" "5557" "5558" "55703" "55811" "55821"
[129] "5631" "5634" "56655" "56953" "56985" "57804" "58497" "6240"
[137] "6241" "64425" "646625" "654364" "661" "7498" "8382" "84172"
[145] "84265" "84284" "84618" "8622" "8654" "87178" "8833" "9060"
[153] "9061" "93034" "953" "9533" "954" "955" "956" "957"
[161] "9583" "9615"
```

Prepare input: named vector of fold changes

```
foldchanges <- res$log2FoldChange
names(foldchanges) <- res$entrez
head(foldchanges)
```

```
      1266      54855      1465      2034      2150      6659
-2.422719  3.201955 -2.313738 -1.888019  3.344508  2.392288
```

Run gage pathway analysis

```
keggres <- gage(foldchanges, gsets = kegg.sets.hs)
attributes(keggres)
```

```
$names
[1] "greater" "less" "stats"
```

Look at top down-regulated pathways

```
head(keggres$less)
```

	p.geomean	stat.mean	p.val
hsa04110 Cell cycle	1.003993e-05	-4.353454	1.003993e-05
hsa03030 DNA replication	8.909558e-05	-3.968611	8.909558e-05
hsa03013 RNA transport	1.470985e-03	-3.007794	1.470985e-03
hsa04114 Oocyte meiosis	1.946905e-03	-2.921710	1.946905e-03
hsa03440 Homologous recombination	2.941989e-03	-2.868141	2.941989e-03
hsa00010 Glycolysis / Gluconeogenesis	6.059196e-03	-2.558327	6.059196e-03

	q.val	set.size	exp1
hsa04110 Cell cycle	0.001606390	120	1.003993e-05
hsa03030 DNA replication	0.007127646	36	8.909558e-05

hsa03013	RNA transport	0.077876201	143	1.470985e-03
hsa04114	Oocyte meiosis	0.077876201	99	1.946905e-03
hsa03440	Homologous recombination	0.094143663	28	2.941989e-03
hsa00010	Glycolysis / Gluconeogenesis	0.161578551	48	6.059196e-03

Generate pathview plots

```
# Cell cycle pathway
pathview(gene.data = foldchanges, pathway.id = "hsa04110")
```

'select()' returned 1:1 mapping between keys and columns

Info: Working in directory C:/Users/HTani/OneDrive/Documents/UCSD23-27/bimm_143/class13

Info: Writing image file hsa04110.pathview.png

```
# PDF version
pathview(gene.data = foldchanges, pathway.id = "hsa04110", kegg.native = FALSE)
```

'select()' returned 1:1 mapping between keys and columns

Warning: reconcile groups sharing member nodes!

```
      [,1] [,2]
[1,] "9"  "300"
[2,] "9"  "306"
```

Info: Working in directory C:/Users/HTani/OneDrive/Documents/UCSD23-27/bimm_143/class13

Info: Writing image file hsa04110.pathview.pdf

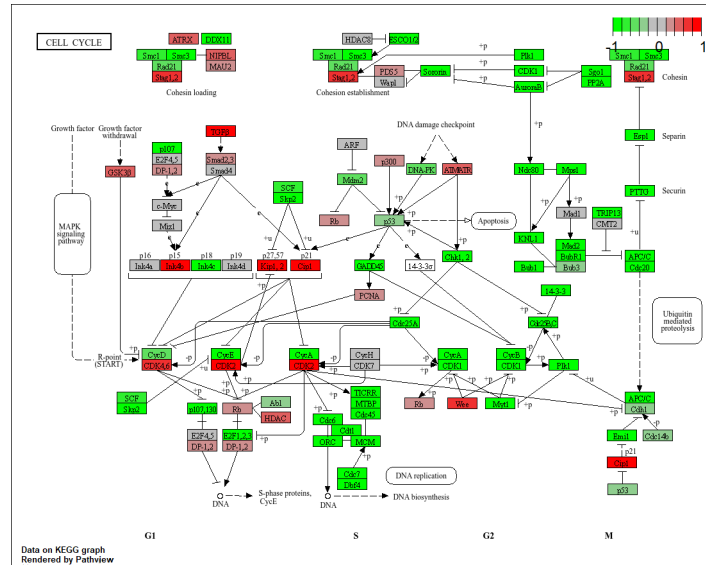


Figure 1: cell cycle pathway

Plot top 5 upregulated pathways

```
keggrespathways <- rownames(keggres$greater)[1:5]
keggresids <- substr(keggrespathways, start = 1, stop = 8)
keggresids
```

```
[1] "hsa04640" "hsa04630" "hsa04142" "hsa00140" "hsa04740"
```

```
pathview(gene.data = foldchanges, pathway.id = keggresids, species = "hsa")
```

'select()' returned 1:1 mapping between keys and columns

Info: Working in directory C:/Users/HTani/OneDrive/Documents/UCSD23-27/bimm_143/class13

Info: Writing image file hsa04640.pathview.png

'select()' returned 1:1 mapping between keys and columns

Info: Working in directory C:/Users/HTani/OneDrive/Documents/UCSD23-27/bimm_143/class13

Info: Writing image file hsa04630.pathview.png

'select()' returned 1:1 mapping between keys and columns

Info: Working in directory C:/Users/HTani/OneDrive/Documents/UCSD23-27/bimm_143/class13

Info: Writing image file hsa04142.pathview.png

'select()' returned 1:1 mapping between keys and columns

Info: Working in directory C:/Users/HTani/OneDrive/Documents/UCSD23-27/bimm_143/class13

Info: Writing image file hsa00140.pathview.png

'select()' returned 1:1 mapping between keys and columns

Info: Working in directory C:/Users/HTani/OneDrive/Documents/UCSD23-27/bimm_143/class13

Info: Writing image file hsa04740.pathview.png

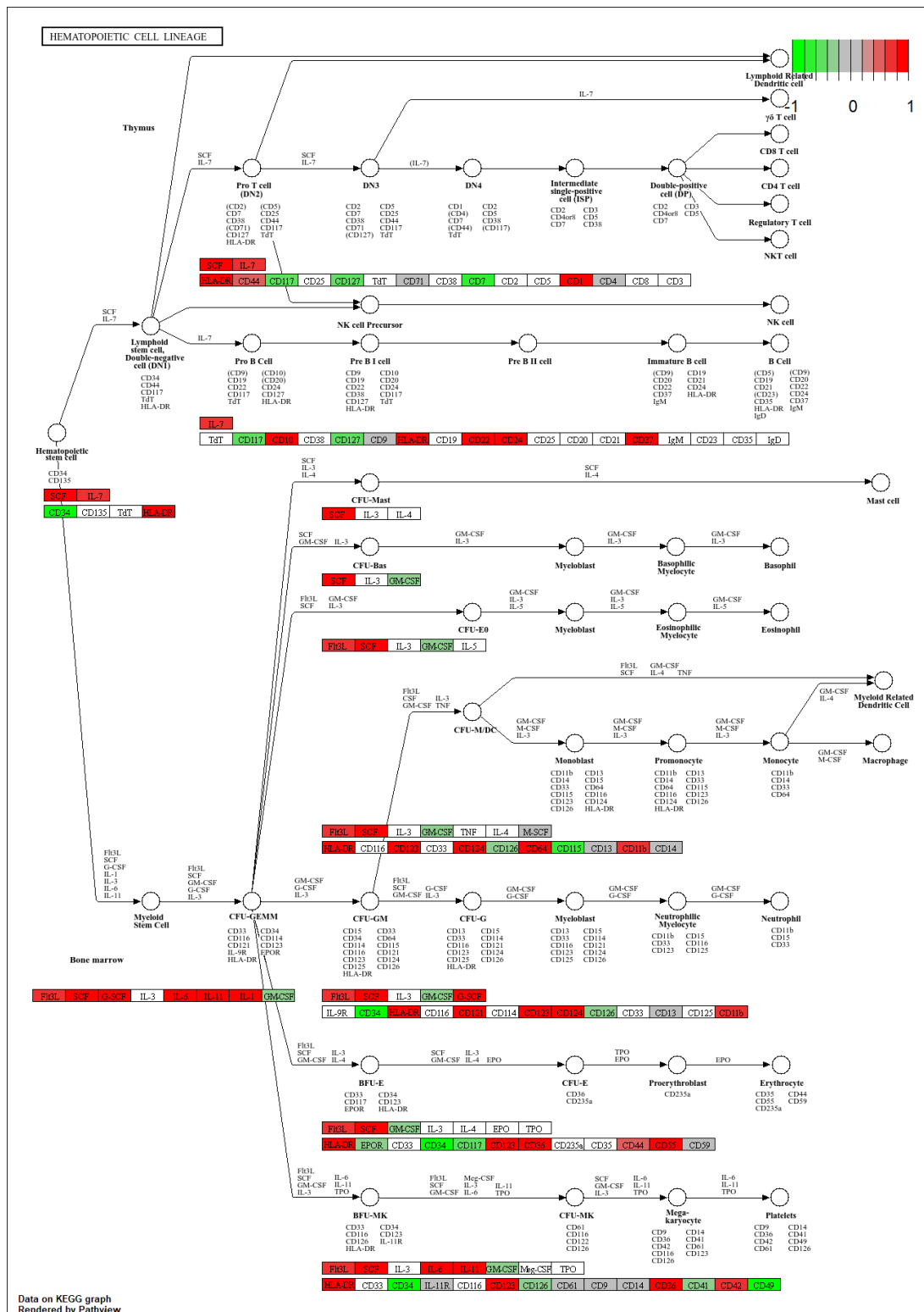


Figure 2: Hematopoietic Cell Lineage

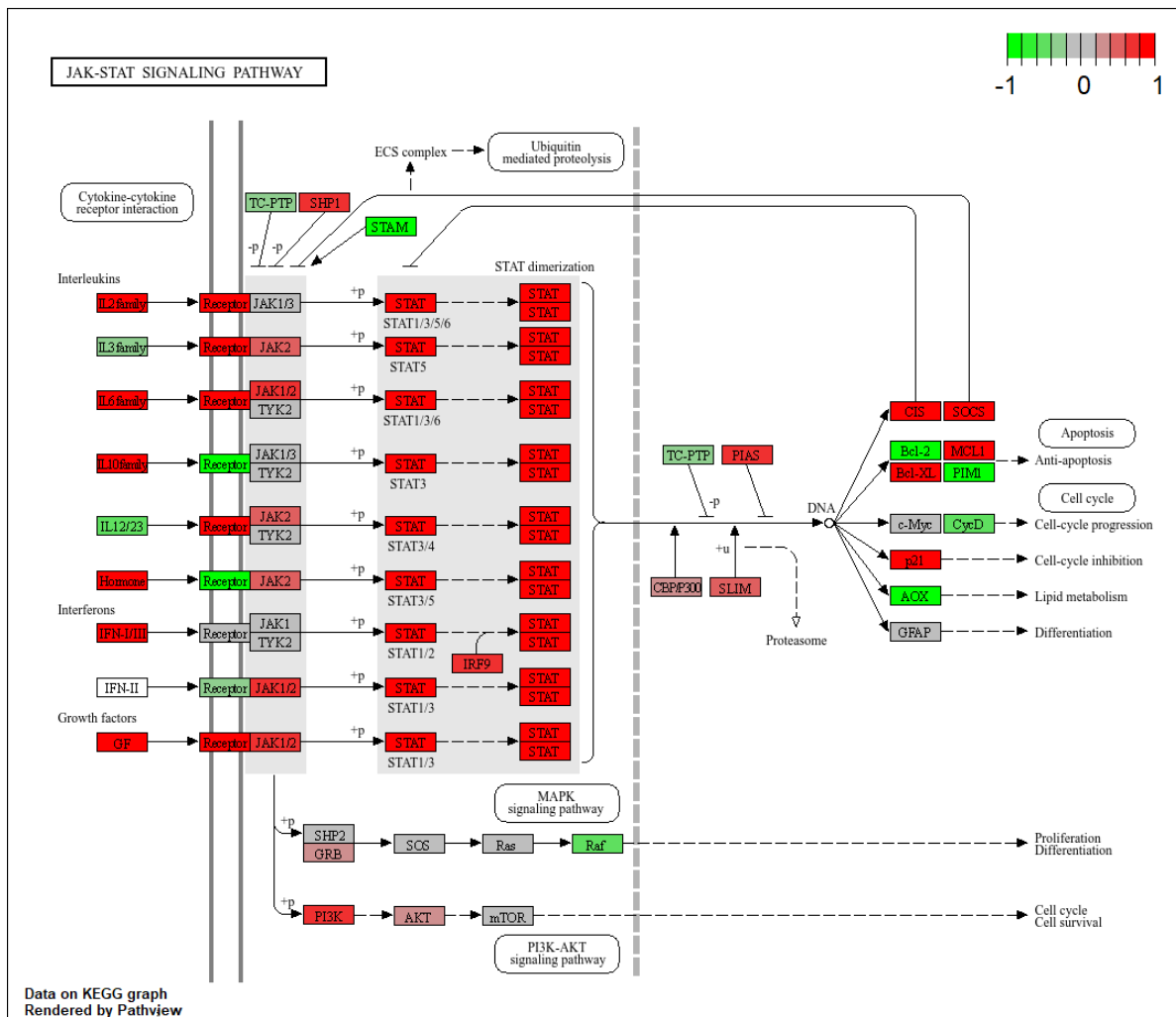


Figure 3: JAK-STAT Signaling Pathway

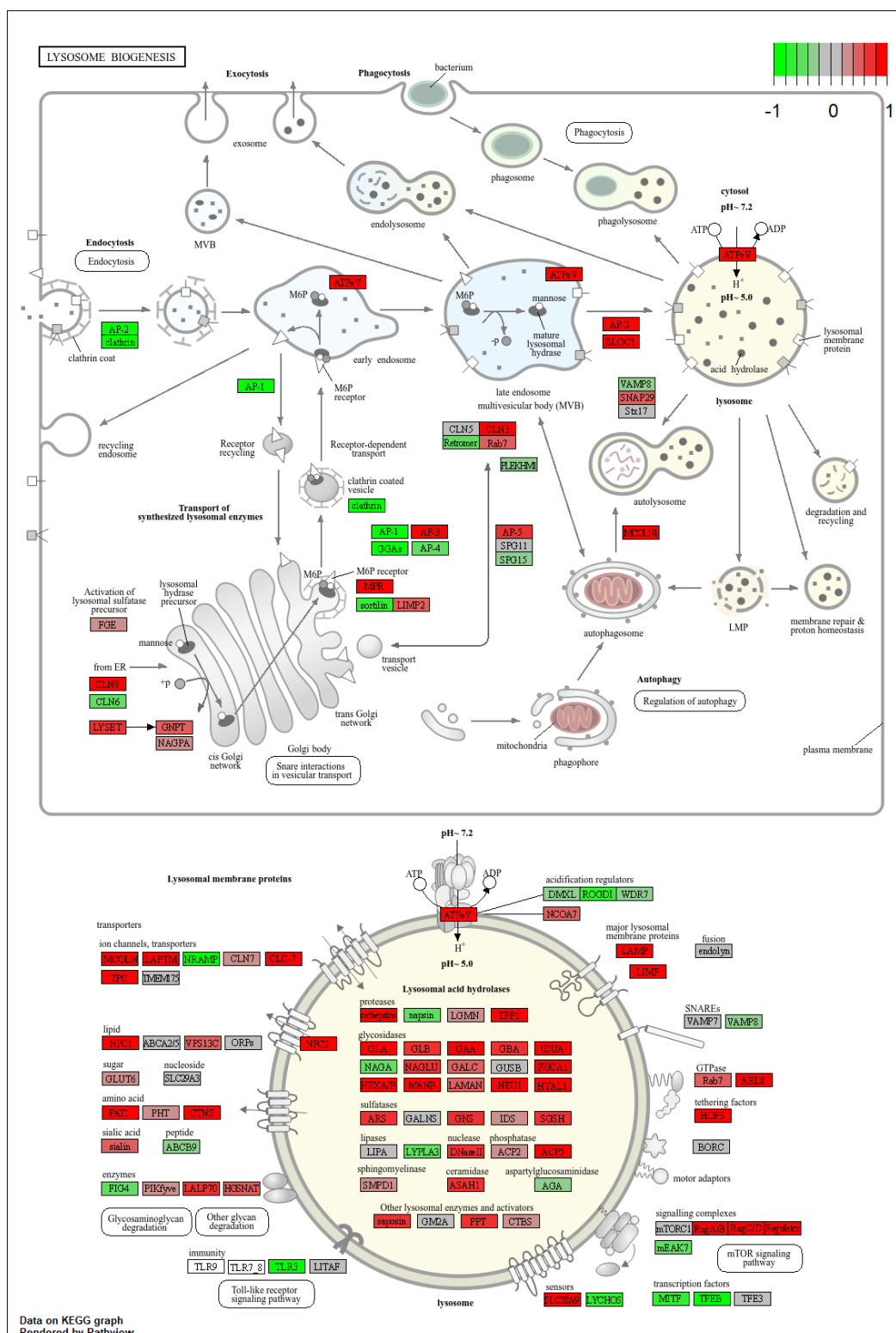
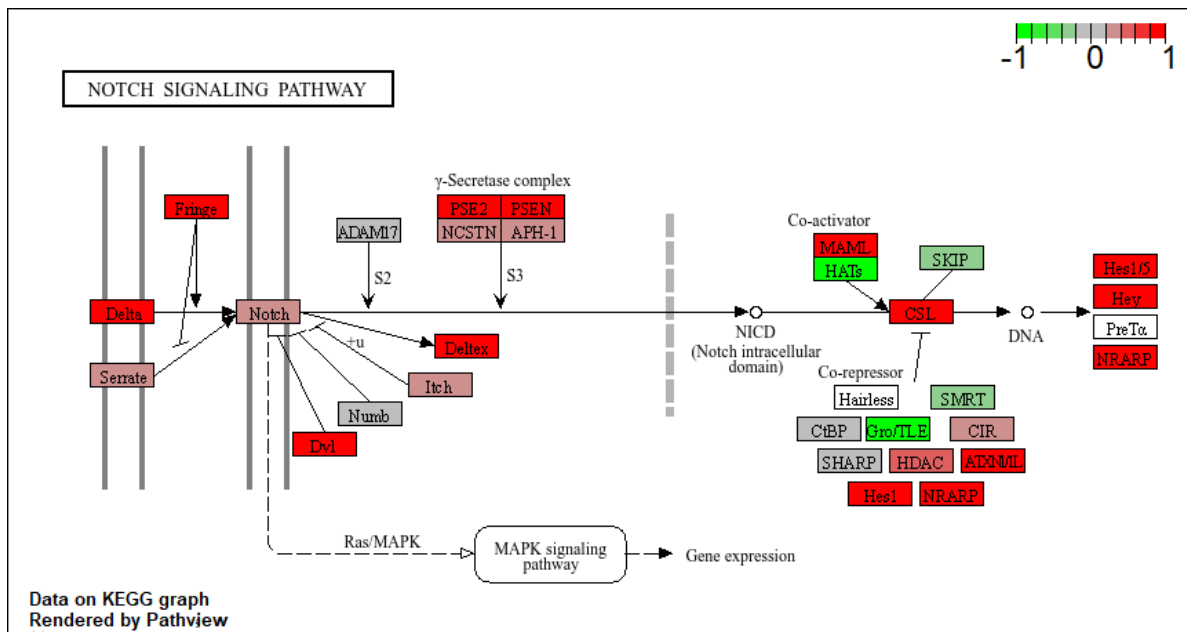


Figure 5: Lysosome



Q. Plot top 5 down-regulated pathways

Same procedure but with `keggres$less`:

Q. Can you do the same procedure as above to plot the pathview figures for the top 5 down-regulated pathways?

```
keggrespathways_down <- rownames(keggres$less)[1:5]
keggresids_down <- substr(keggrespathways_down, start = 1, stop = 8)
keggresids_down
```

```
[1] "hsa04110" "hsa03030" "hsa03013" "hsa04114" "hsa03440"
```

```
pathview(gene.data = foldchanges, pathway.id = keggresids_down, species = "hsa")
```

'select()' returned 1:1 mapping between keys and columns

Info: Working in directory C:/Users/HTani/OneDrive/Documents/UCSD23-27/bimm_143/class13

Info: Writing image file hsa04110.pathview.png

'select()' returned 1:1 mapping between keys and columns

Info: Working in directory C:/Users/HTani/OneDrive/Documents/UCSD23-27/bimm_143/class13

Info: Writing image file hsa03030.pathview.png

'select()' returned 1:1 mapping between keys and columns

Info: Working in directory C:/Users/HTani/OneDrive/Documents/UCSD23-27/bimm_143/class13

Info: Writing image file hsa03013.pathview.png

'select()' returned 1:1 mapping between keys and columns

Info: Working in directory C:/Users/HTani/OneDrive/Documents/UCSD23-27/bimm_143/class13

Info: Writing image file hsa04114.pathview.png

'select()' returned 1:1 mapping between keys and columns

Info: Working in directory C:/Users/HTani/OneDrive/Documents/UCSD23-27/bimm_143/class13

Info: Writing image file hsa03440.pathview.png

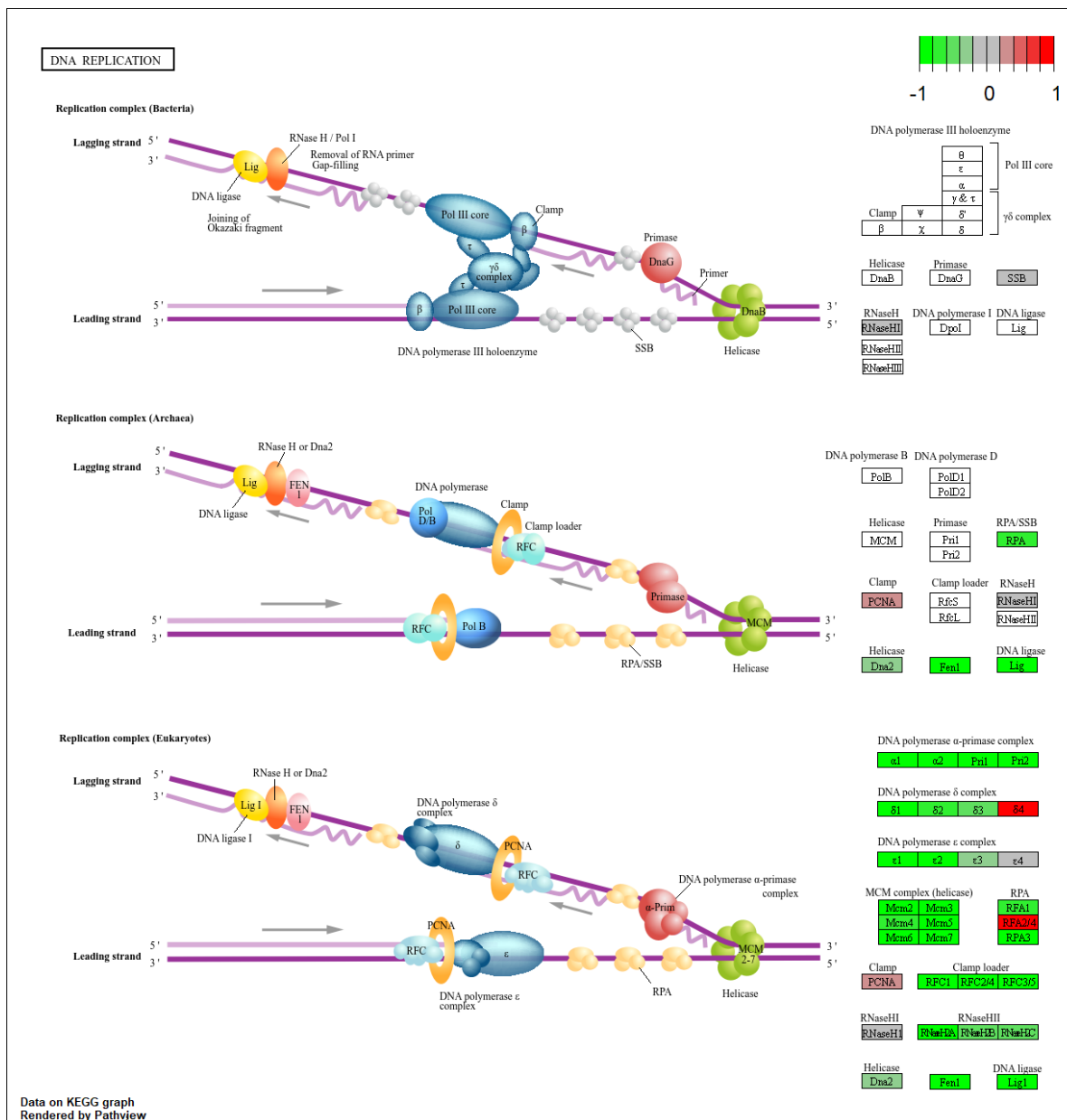


Figure 7: DNA Replication

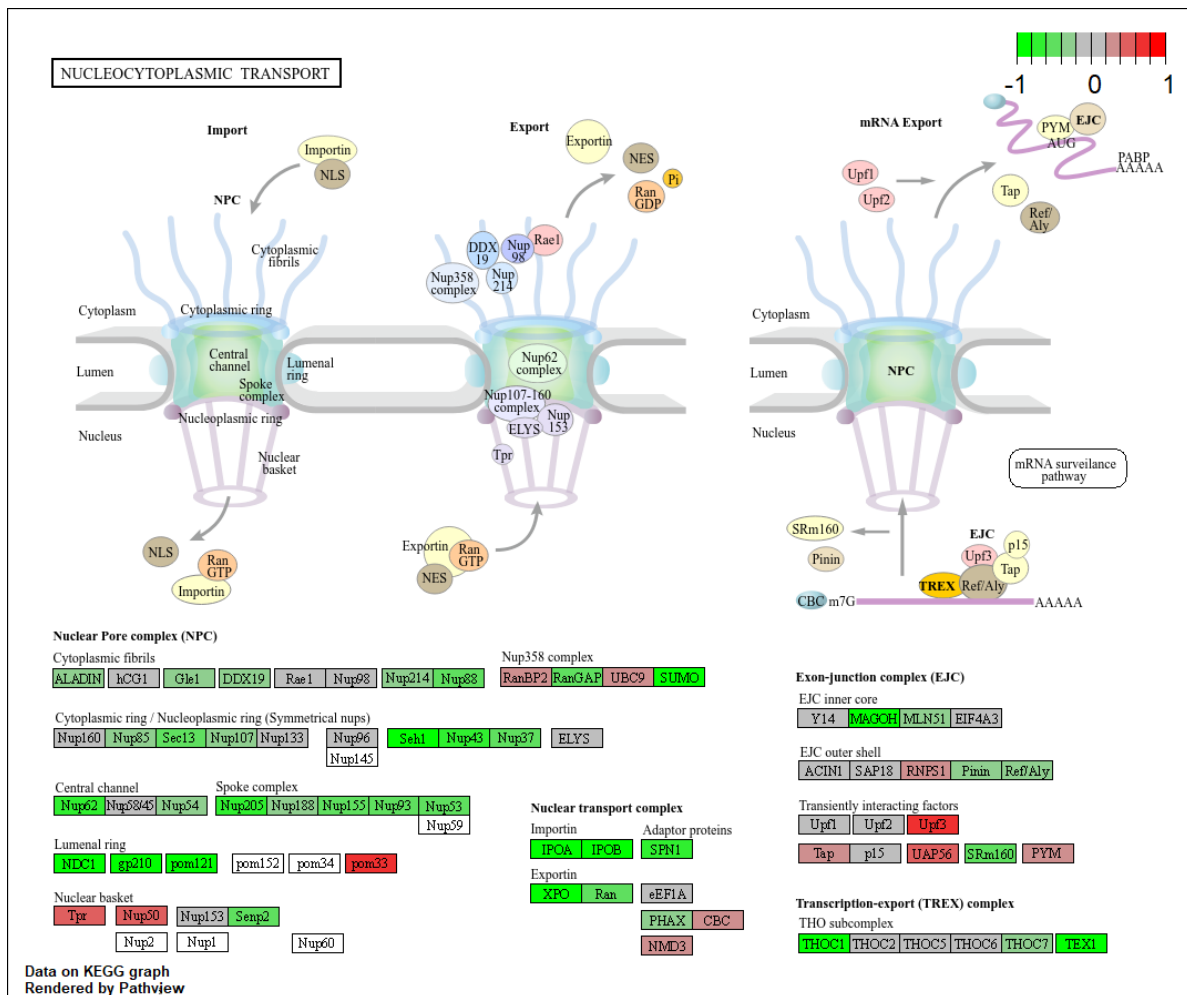


Figure 8: RNA Transport

G0:0002009	morphogenesis of an epithelium	3.227439e-04	3.429317	3.227439e-04
G0:0016337	cell-cell adhesion	8.195506e-04	3.163057	8.195506e-04
		q.val	set.size	exp1
G0:0007156	homophilic cell adhesion	0.1337100	107	4.892477e-05
G0:0060429	epithelium development	0.1337100	478	6.727546e-05
G0:0007610	behavior	0.2299150	402	1.735208e-04
G0:0048729	tissue morphogenesis	0.2455518	403	2.470962e-04
G0:0002009	morphogenesis of an epithelium	0.2565814	326	3.227439e-04
G0:0016337	cell-cell adhesion	0.3656453	318	8.195506e-04

\$less

		p.geomean	stat.mean	p.val
G0:0000279	M phase	1.475361e-16	-8.323749	1.475361e-16
G0:0048285	organelle fission	7.498413e-16	-8.160305	7.498413e-16
G0:0000280	nuclear division	2.135098e-15	-8.034814	2.135098e-15
G0:0007067	mitosis	2.135098e-15	-8.034814	2.135098e-15
G0:0000087	M phase of mitotic cell cycle	5.927567e-15	-7.891758	5.927567e-15
G0:0007059	chromosome segregation	1.055918e-11	-6.988373	1.055918e-11
		q.val	set.size	exp1
G0:0000279	M phase	5.864560e-13	492	1.475361e-16
G0:0048285	organelle fission	1.490309e-12	373	7.498413e-16
G0:0000280	nuclear division	2.121754e-12	349	2.135098e-15
G0:0007067	mitosis	2.121754e-12	349	2.135098e-15
G0:0000087	M phase of mitotic cell cycle	4.712416e-12	359	5.927567e-15
G0:0007059	chromosome segregation	6.995458e-09	141	1.055918e-11

\$stats

	stat.mean	exp1
G0:0007156	homophilic cell adhesion	3.971899
G0:0060429	epithelium development	3.834595
G0:0007610	behavior	3.594057
G0:0048729	tissue morphogenesis	3.498983
G0:0002009	morphogenesis of an epithelium	3.429317
G0:0016337	cell-cell adhesion	3.163057

Section 4. Reactome Analysis

```
sig_genes <- res[res$padj <= 0.05 & !is.na(res$padj), "symbol"]
print(paste("Total number of significant genes:", length(sig_genes)))
```

```
[1] "Total number of significant genes: 8149"
```

```
write.table(sig_genes, file = "significant_genes.txt",
            row.names = FALSE, col.names = FALSE, quote = FALSE)
```

Q: What pathway has the most significant “Entities p-value”? Do the most significant pathways listed match your previous KEGG results? What factors could cause differences between the two methods?

Cell Cycle (R-HSA-1640170) has the most significant Entities p-value at 2.68e-05, with 493 out of 729 entities found. The second most significant is Cell Cycle, Mitotic (R-HSA-69278) at $p = 2.83e-05$.

Yes, they match very well. In the KEGG/gage analysis, the top downregulated pathway was hsa04110 Cell Cycle ($p = 1e-05$), and here in Reactome the top hit is also Cell Cycle. Several other overlapping themes appear in both analyses: DNA replication, cell cycle checkpoints, and mitotic processes which are all consistent with HOXA1’s known role in cell cycle progression.

The factors that lead to the differences might be different pathway databases, statistical approaches, ID mapping and gene universes and multiple testing corrections.

Section 5. GO online (OPTIONAL)

Q: What pathway has the most significant “Entities p-value”? Do the most significant pathways listed match your previous KEGG results? What factors could cause differences between the two methods?

The most significant biological process terms by raw p-value are broad umbrella terms like metabolic process ($p = 2.61e-135$) and cellular process ($p = 8.51e-134$). Looking at specific and biologically meaningful terms, cell cycle-related processes dominate: regulation of cell cycle (GO:0051726, $p = 1.38e-50$), mitotic cell cycle (GO:0000278, $p = 5.05e-50$), mitotic cell cycle process (GO:1903047, $p = 1.97e-48$), and chromosome segregation (GO:0007059, $p = 3.56e-22$).

Yes, these results strongly match the previous KEGG analysis. KEGG identified Cell Cycle (hsa04110) as the top downregulated pathway, Reactome also ranked Cell Cycle first ($p = 2.68e-05$), and here the GO analysis confirms that mitotic cell cycle, cell cycle regulation, DNA replication, and chromosome segregation are all highly enriched. All three methods converge on the same conclusion: loss of HOXA1 predominantly disrupts cell cycle and mitotic processes.

The factors that lead to the differences might be GO uses a hierarchical ontology and KEGG defines flat, discrete pathway maps with interaction information. Also statistical approaches differ and GO has broader

annotation coverage than KEGG, so it captures more general biological processes. GO results contain substantial redundancy (e.g., “mitotic cell cycle” and “mitotic cell cycle process” share nearly the same genes), which does not occur in KEGG. The different multiple testing correction procedures and different numbers of tested categories affect which results pass significance thresholds.